

AgentHOI: Towards Multi-Agent Collaborative Human-Object Interaction Detection

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Abstract—Human-object interaction (HOI) detection is an important task in computer vision. Existing methods primarily rely on end-to-end deep learning paradigms with specifically designed architectures. With the rapid advancement of large language models (LLMs) and agent technologies, incorporating agent-based collaborative mechanisms into visual understanding tasks has emerged as a promising research direction. In this article, we propose AgentHOI, which, to the best of our knowledge, represents an early attempt to apply multi-agent debate mechanisms specifically to HOI detection. Our framework comprises six collaboratively operating agents: the Hypothesis Proposer (HP) generates initial interaction hypotheses based on vision-language models; the Visual Analysis Agent (VAA) extracts spatial relationships and contact evidence; the Semantic Expert Agent (SEA) leverages LLMs for commonsense reasoning; the Physics Auditor Agent (PAA) validates the physical plausibility of interactions; the Debate Moderator (DM) orchestrates multi-round debates to facilitate viewpoint exchange; and the Consensus Arbiter (CA) synthesizes evidence from all parties to render final verdicts. Through this novel "hypothesis-debate-consensus" paradigm, multiple agents analyze and reason about interaction hypotheses from visual, semantic, and physical perspectives. We conduct experiments on the HICO-DET benchmark and provide a detailed analysis of the framework's characteristics and limitations. Code is available at <https://drliuqi.github.io/>

Index Terms—Agent, human-object interaction (HOI), scene understanding.

I. INTRODUCTION

IN THE field of computer vision [1], [2], [3], human-object interaction (HOI) detection aims to identify the interaction relationships between humans and objects in images, producing outputs in the form of ⟨human, action, object⟩ triplets [4], [5], [6], [7]. As a bridge connecting object detection and high-level visual understanding [8], [9], HOI detection has broad applications in action recognition, human behavior analysis, robotic manipulation, and image captioning [10], [11], [12], [13], [14], [15], [16], [17].

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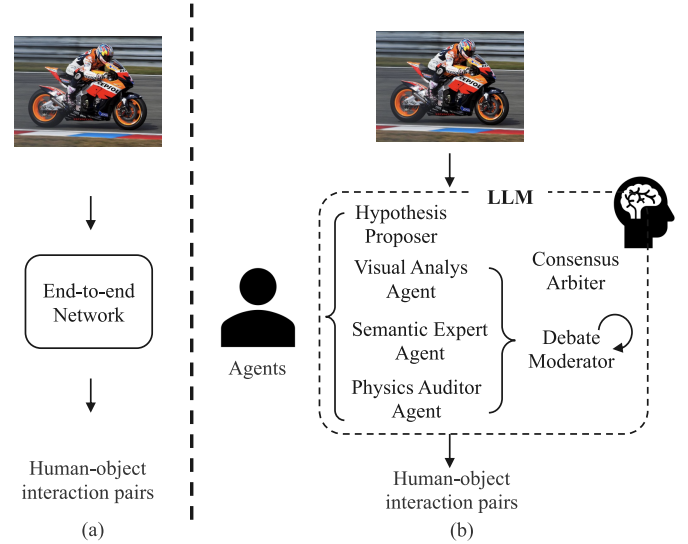


Fig. 1. Comparison of two different HOI frameworks. (a) *Traditional HOI networks*: It completes the task by training and updating deep learning networks. (b) *AgentHOI*: It employs multiple collaborative agents to perform HOI detection, with each agent responsible for a distinct aspect of the reasoning process.

In recent years, the field of HOI detection has witnessed significant progress. Early two-stage methods first employ object detectors to obtain candidate bounding boxes of humans and objects, then design interaction classification networks to determine interaction types [18], [19], [20]. With the success of DETR [21], [22], [23], Transformer-based one-stage methods have gradually become mainstream. These approaches directly predict HOI triplets through carefully designed interaction queries, achieving remarkable performance improvements [23], [24], [25], [26]. Additionally, some works have attempted to incorporate auxiliary information such as human pose and vision-language pretrained models [7], [23] to enhance interaction understanding. Despite continuous performance breakthroughs, these methods essentially follow the same paradigm: designing end-to-end neural networks that learn direct mappings from images to interaction labels through large-scale data-driven training [as shown in Fig. 1(a)].

The field of artificial intelligence is undergoing a paradigm shift driven by LLMs [27]. In particular, LLM-based AI agents have demonstrated powerful capabilities in reasoning, planning, and tool utilization, achieving remarkable results in code generation, scientific research, and complex decision-making

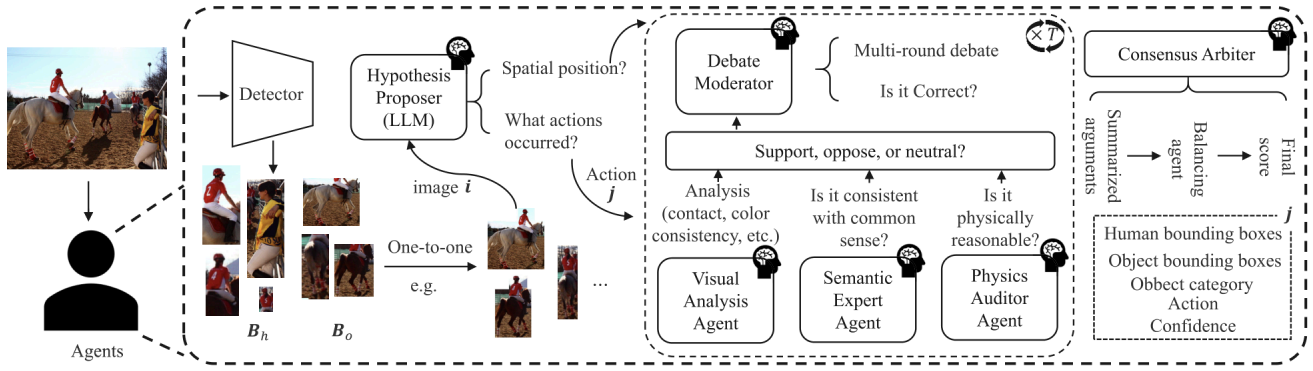


Fig. 2. Overview of the proposed AgentHOI framework. The framework consists of four stages: 1) object detection to identify humans and objects; 2) Hypothesis generation by the HP using vision-language models; 3) Multi-Agent debate where three analyst agents (VAA, SEA, PAA) evaluate hypotheses from visual, semantic, and physical perspectives through multiple rounds moderated by the DM; and 4) Consensus formation where the CA renders final verdicts through weighted voting.

tasks [28]. Furthermore, multi-agent systems, by enabling multiple agents to collaborate or debate, can integrate knowledge from different perspectives to produce more comprehensive and reliable outputs [29], [30], [31]. These advances inspire us to consider: can we introduce multi-agent collaborative mechanisms into visual understanding tasks to bring new possibilities for HOI detection?

Based on the above considerations, we propose AgentHOI, a multi-agent debate framework for HOI detection [as shown in Fig. 1(b)]. This represents an early attempt to apply multi-agent technologies to the HOI detection task. Specifically, we design six agents with distinct expertise: Hypothesis Proposer (HP) generates initial interaction hypotheses using vision-language models; Visual Analysis Agent (VAA) extracts spatial relationships and contact evidence from images; Semantic Expert Agent (SEA) leverages LLMs for commonsense reasoning; Physical Auditor Agent (PAA) validates the physical plausibility of interactions; Debate Moderator (DM) orchestrates multi-round debates to facilitate viewpoint exchange; Consensus Arbiter (CA) synthesizes opinions from all parties to render final verdicts. Through the “hypothesis-debate-consensus” workflow, these agents thoroughly analyze and reason about interaction hypotheses from visual, semantic, and physical dimensions.

II. RELATED WORK

A. HOI Detection

Early two-stage methods follow a detect-then-classify paradigm: they first employ object detectors to obtain human and object proposals, then design interaction classification networks. Representative works include HICO-DET [32], which established the benchmark, iCAN [33] with instance-centric attention, and TIN [34] leveraging interactiveness knowledge. Inspired by DETR, one-stage Transformer-based methods have become the dominant paradigm. QPIC [35] extends DETR with interaction queries for end-to-end HOI prediction. Recently, vision-language pretrained models have been leveraged to further improve performance.

B. AI Agents

An AI agent is an autonomous entity that perceives its environment, makes decisions, and takes actions to achieve

goals [27]. The emergence of LLMs has revolutionized this field. Modern LLM-based agents leverage foundation models such as GPT-4 [36] and LLaMA [37] for powerful reasoning, planning, and tool utilization capabilities [28]. A typical agent consists of four components: the LLM as the brain for decision-making, perception modules for multimodal input processing, memory for storing historical information, and action modules for tool usage and environment interaction [38].

III. METHOD

A. Framework Overview

Given an input image $I \in \mathbb{R}^{H \times W \times 3}$, the goal of HOI detection is to identify all HOI instances, where each instance is represented as a triplet $\langle b_h, b_o, c_o, v \rangle$, consisting of the human bounding box b_h , the object bounding box b_o , the object category c_o , and the interaction verb v . Unlike traditional end-to-end methods that directly learn a mapping $f: y \rightarrow \{(b_h, b_o, c_o, v)\}$, our training-free AgentHOI framework formulates HOI detection as a multi-agent deliberative process. As illustrated in Fig. 2, the framework consists of six collaborative agents organized in a “hypothesis-debate-consensus” pipeline.

Formally, let $\mathcal{H} = \{h_1, h_2, \dots, h_N\}$ denote the set of initial HOI hypotheses. Each hypothesis h_i undergoes T rounds of debate, producing a sequence of arguments $\mathcal{A}_i = \{\mathcal{A}_i^{(1)}, \mathcal{A}_i^{(2)}, \dots, \mathcal{A}_i^{(T)}\}$, where $\mathcal{A}_i^{(t)}$ represents the arguments from all agents in round t . The final output is determined by the following equation:

$$\mathcal{Y} = \text{Arbiter}(\{(h_i, \mathcal{A}_i) \mid h_i \in \mathcal{H}\}). \quad (1)$$

B. Object Detection

The first stage of our framework is to detect all humans and objects in the input image. We employ a pretrained object detector $\mathcal{D}$ to obtain

$$\mathcal{B}_h, \mathcal{B}_o = \mathcal{D}(I) \quad (2)$$

where $\mathcal{B}_h = \{(b_h^i, s_h^i)\}_{i=1}^M$ denotes the set of detected humans with bounding boxes and confidence scores, and $\mathcal{B}_o = \{(b_o^j, c_o^j, s_o^j)\}_{j=1}^K$ denotes the detected objects with their bounding boxes, categories, and confidence scores.

In our implementation, we use YOLOv8 for efficient detection. Each bounding box is represented as $\mathbf{b} = [x_1, y_1, x_2, y_2]$, where (x_1, y_1) and (x_2, y_2) are the top-left and bottom-right coordinates, respectively.

C. Hypothesis Proposer

The HP is responsible for generating initial HOI hypotheses by pairing detected humans with objects and predicting potential interactions. This agent leverages vision-language models to understand the scene context and propose plausible interactions.

1) *Spatial Relation Computation*: For each human-object pair $(\mathbf{b}_h, \mathbf{b}_o)$, we compute a comprehensive spatial relation feature. First, we compute the center coordinates via

$$(c_h^x, c_h^y) = \left(\frac{x_1^h + x_2^h}{2}, \frac{y_1^h + y_2^h}{2} \right) \quad (3)$$

$$(c_o^x, c_o^y) = \left(\frac{x_1^o + x_2^o}{2}, \frac{y_1^o + y_2^o}{2} \right). \quad (4)$$

The spatial relation is characterized by several geometric features. The relative position between the human and object is defined as the displacement of the object center from the human center

$$(\Delta x, \Delta y) = (c_o^x - c_h^x, c_o^y - c_h^y). \quad (5)$$

The Euclidean distance between them is computed as follows:

$$d = \sqrt{\Delta x^2 + \Delta y^2}. \quad (6)$$

To capture the overlap between bounding boxes, we employ the intersection over union (IoU) metric

$$\text{IoU}(\mathbf{b}_h, \mathbf{b}_o) = \frac{|\mathbf{b}_h \cap \mathbf{b}_o|}{|\mathbf{b}_h \cup \mathbf{b}_o|}. \quad (7)$$

Additionally, we compute the relative position and vertical relation between the human and object bounding boxes based on their center coordinates. The relative position indicates whether the human is located to the left or right of the object

$$\text{relative_position} = \begin{cases} \text{left,} & \text{if } c_h^x < c_o^x \\ \text{right,} & \text{otherwise.} \end{cases} \quad (8)$$

Similarly, the vertical relation captures whether the human is positioned above or below the object

$$\text{vertical_relation} = \begin{cases} \text{above,} & \text{if } c_h^y < c_o^y \\ \text{below,} & \text{otherwise.} \end{cases} \quad (9)$$

These discrete spatial descriptors provide intuitive categorical representations of the human-object spatial configuration, complementing the continuous geometric features.

2) *Interaction Prediction*: For each human-object pair, we use the BLIP Image-Text Matching (ITM) model to score potential interactions. Given the object category c_o and a candidate verb v from the verb vocabulary $\mathcal{V}$, we construct a text prompt

$$\mathbf{p}_{v,c_o} = \text{"a person } v \text{ a } c_o\text{"}. \quad (10)$$

We crop the interaction region $\mathbf{I}_{\text{crop}}$ that encompasses both the human and object. The crop boundaries are determined by the union of their bounding boxes

$$\begin{aligned} x_{\min} &= \min(x_1^h, x_1^o), & x_{\max} &= \max(x_2^h, x_2^o) \\ y_{\min} &= \min(y_1^h, y_1^o), & y_{\max} &= \max(y_2^h, y_2^o) \end{aligned} \quad (11)$$

$$\mathbf{I}_{\text{crop}} = \mathbf{I}[y_{\min} : y_{\max}, x_{\min} : x_{\max}]. \quad (12)$$

The interaction score is computed via BLIP-ITM

$$s_{v,c_o} = \text{BLIP}_{\text{ITM}}(\mathbf{I}_{\text{crop}}, \mathbf{p}_{v,c_o}). \quad (13)$$

To reduce computational cost, we first select a subset of relevant verbs $\mathcal{V}_{c_o} \subset \mathcal{V}$ based on the object category using predefined affordance mappings. The affordance mappings are constructed through a combination of two sources: 1) commonsense knowledge generated by querying an LLM about which actions are applicable to each object category; and 2) manual verification and refinement based on the dataset annotation statistics to ensure coverage. For example, the object category "bicycle" is mapped to verbs such as {ride, hold, push, repair, sit_on,}. The top- k interactions with the highest scores are retained as hypotheses

$$\mathcal{H}_{h,o} = \text{Top-}k(\{(v, s_{v,c_o}) \mid v \in \mathcal{V}_{c_o}\}). \quad (14)$$

Each hypothesis $h \in \mathcal{H}$ is formally represented as follows:

$$\mathbf{h} = (\mathbf{b}_h, \mathbf{b}_o, c_o, v, s_{\text{init}}) \quad (15)$$

where s_{init} is the initial confidence score from the proposer.

D. Visual Analysis Agent

The VAA examines the visual evidence in the image to assess whether the proposed interaction is visually supported. This agent focuses on four aspects: contact analysis, spatial compatibility, color consistency, and motion cues.

1) *Contact Analysis*: Physical contact is crucial for many interactions (e.g., "hold," "carry," "ride"). We analyze the contact relationship between human and object bounding boxes

$$\text{Area}_{\text{overlap}} = \max(0, x_2^o - x_1^o) \times \max(0, y_2^o - y_1^o) \quad (16)$$

where $(x_1^o, y_1^o) = (\max(x_1^h, x_1^o), \max(y_1^h, y_1^o))$ and $(x_2^o, y_2^o) = (\min(x_2^h, x_2^o), \min(y_2^h, y_2^o))$. The contact score is computed as follows:

$$s_{\text{contact}} = \begin{cases} \min\left(1.0, \frac{2 \cdot \text{Area}_{\text{overlap}}}{\text{Area}(\mathbf{b}_h)}\right), & \text{if } \text{IoU}(\mathbf{b}_h, \mathbf{b}_o) > 0 \\ \frac{1}{2} \left(1 - \frac{d}{d_{\max}}\right), & \text{otherwise} \end{cases} \quad (17)$$

where $d_{\max} = (H^2 + W^2)^{1/2}$ is the image diagonal length.

2) *Spatial Compatibility*: We evaluate the spatial relationship based on the interaction category. Interactions are classified into three types: contact-required (Actions like "hold," "carry," "wear" require physical proximity); distance-required (Actions like "watch," "point," "throw" typically involve separation); and flexible-distance (Actions like "walk," "run" are less constrained by distance). The spatial compatibility score is

$$s_{\text{spatial}} = \exp\left(-\frac{d}{\sigma_v}\right) \quad (18)$$

where σ_v is a verb-specific scale parameter (smaller for contact-required actions, larger for distance-required actions).

3) *Color Consistency*: We compute color histogram similarity between the human and object regions to detect potential occlusion or interaction artifacts

$$s_{\text{color}} = \frac{1 + \rho(\text{Hist}(\mathbf{I}_{b_h}), \text{Hist}(\mathbf{I}_{b_o}))}{2} \quad (19)$$

where $\rho(\cdot, \cdot)$ denotes histogram correlation and $\mathbf{I}_b$ is the image region within bounding box b .

4) *Action-Specific Analysis*: The visual evidence score integrates all components with action-specific weights. Let $s_b = s_{\text{base}}$, $s_c = s_{\text{contact}}$, and $s_p = s_{\text{spatial}}$ for brevity

$$s_{\text{AC}} = \begin{cases} 0.3 s_b + 0.7 s_c, & \text{if } v \in \mathcal{V}_{\text{contact}} \\ 0.3 s_b + 0.3 (1 - s_c) + 0.4 s_p, & \text{if } v \in \mathcal{V}_{\text{distance}} \\ 0.4 s_b + 0.6 s_p, & \text{otherwise} \end{cases} \quad (20)$$

where $s_b = s_{\text{base}}$ is a plausibility score output by the rule generator, which refers to a score obtained by applying an LLM to evaluate the rationality of the input image with respect to the linguistic description and physical constraints. s_c is the contact score, and s_p is the spatial compatibility score. The final score is computed by

$$s_{\text{VAA}} = s_{\text{AC}} * 0.4 + s_{\text{contact}} * 0.3 + s_{\text{spatial}} * 0.2 + s_{\text{color}} * 0.1. \quad (21)$$

Based on the visual evidence score, the VAA agent determines its stance

$$\text{Stance}_{\text{VAA}} = \begin{cases} \text{support}, & \text{if } s_{\text{VAA}} > \theta_{\text{sup}} \\ \text{oppose}, & \text{if } s_{\text{VAA}} < \theta_{\text{opp}} \\ \text{neutral}, & \text{otherwise} \end{cases} \quad (22)$$

where $\theta_{\text{sup}} = 0.6$ and $\theta_{\text{opp}} = 0.4$ are threshold parameters.

E. Semantic Expert Agent

The SEA leverages LLMs to assess the semantic plausibility of proposed interactions based on commonsense knowledge. For each verb-object pair (v, c_o) , we query an LLM to generate interaction rules. The prompt is structured as $\mathcal{P}_{v, c_o}$. (“Analyze the interaction ”v” with object ”c_o”. Return JSON with keys: plausibility (0-1 float), physical_possible (bool), common_sense (bool), safe (bool), reasoning (string <= 60 chars).”) In addition, to ensure efficiency, we maintain a cache of LLM responses

$$\mathcal{R}_{v, c_o} = \begin{cases} \text{Cache}[v, c_o], & \text{if } (v, c_o) \in \text{Cache} \\ \text{LLM}(\mathcal{P}_{v, c_o}), & \text{otherwise.} \end{cases} \quad (23)$$

The final score is obtained from the language model’s rating in $\mathcal{R}_{v, c_o}$, and then we apply (22) to determine whether the hypothesis is supported.

F. Physics Auditor Agent

The Physics Auditor Agent (PAA) validates whether the proposed interaction is physically plausible given the spatial configuration of the human and object. Taking the HICO-Det dataset as an example, we leverage an LLM to categorize all action classes into 16 major categories, including hand manipulation actions (e.g., “hold,” “carry”), foot-related actions (e.g., “kick,” “step on”), sitting/standing/lying actions (e.g., “sit on,” “sit at”), riding actions (e.g., “ride,” “drive”), throwing actions (e.g., “throw”), pushing/pulling actions (e.g., “push,” “pull”), tool-using actions (e.g., “cut with”), eating/drinking actions (e.g., “eat,” “drink with”), wearing actions (e.g., “wear”), cleaning actions (e.g., “wash,” “dry”), body contact actions (e.g., “kiss,” “hug”), operating actions (e.g., “open,” “close”), sports actions (e.g., “jump”), observing actions (e.g., “watch,” “inspect”), pointing actions (e.g., “point”), and other actions (e.g., “stand,” “board,” “exit”). Subsequently, we utilize the relative distance d between the human and object, along with the human center position, object center position, human height, and object height, to verify whether the action category predicted by the LLM is physically plausible, yielding a physical score s_p . The PAA agent integrates the physical analysis with rule-based knowledge

$$s_{\text{PAA}} = \alpha \cdot s_{\text{rule}} + (1 - \alpha) \cdot s_p \quad (24)$$

where s_{rule} denotes the rule-based plausibility score and α is a weighting coefficient.

G. Debate Moderator

The DM orchestrates the multi-round debate process among the three analyst agents.

1) *Debate Protocol*: For each hypothesis h , the debate proceeds for at most T rounds (default $T = 3$). In each round t , every analyst agent $a \in \{\text{VAA}, \text{SEA}, \text{PAA}\}$ produces an argument

$$A_a^{(t)} = (\text{stance}_a^{(t)}, s_a^{(t)}, e_a^{(t)}) \quad (25)$$

where $\text{stance}_a^{(t)} \in \{\text{support}, \text{oppose}, \text{neutral}\}$, $s_a^{(t)}$ is the confidence score and $e_a^{(t)}$ is the evidence.

2) *Argument Exchange*: Each agent can access the arguments from previous rounds, enabling response and rebuttal

$$A_a^{(t)} = \text{Agent}_a(I, h, \{A^{(1)}, \dots, A^{(t-1)}\}). \quad (26)$$

This mechanism allows agents to reinforce their position if supported by other agents, adjust their confidence based on compelling counter-evidence, and respond to specific challenges raised by other agents during the debate process.

3) *Consensus Detection*: After each round, the moderator checks for consensus

$$\text{Consensus}^{(t)} = \frac{\max_{s \in \mathcal{S}} |\{a : \text{stance}_a^{(t)} = s\}|}{|\mathcal{A}|} \geq \theta_{\text{cons}} \quad (27)$$

where $\mathcal{S} = \{\text{support}, \text{oppose}, \text{neutral}\}$ and θ_{cons} is the consensus threshold. If consensus is reached, the debate terminates early. Otherwise, it continues until the maximum number of rounds.

H. Consensus Arbiter

The CA serves as the final decision-maker by aggregating all arguments from the debate process. Each agent is assigned a weight reflecting its expertise, defined as $\mathbf{w} = (w_{VAA}, w_{SEA}, w_{PAA}) = (0.35, 0.35, 0.30)$. The CA computes a weighted score by combining each agent's stance and confidence

$$s_{\text{weighted}} = \sum_a w_a \cdot g(\text{stance}_a, s_a) \quad (28)$$

where the stance-to-score function $g(\cdot)$ converts qualitative stances into numerical contributions

$$g(\text{stance}, s) = \begin{cases} s, & \text{if stance} = \text{support} \\ 0.5 \cdot (1 - s), & \text{if stance} = \text{oppose} \\ 0.5, & \text{if stance} = \text{neutral}. \end{cases} \quad (29)$$

In addition to weighted scores, the CA counts the votes for each stance across all agents and debate rounds. Let $n_{\text{sup}} = \sum_{a,t} \mathbb{I}[\text{stance}_a^{(t)} = \text{support}]$ and $n_{\text{opp}} = \sum_{a,t} \mathbb{I}[\text{stance}_a^{(t)} = \text{oppose}]$ denote the total number of supporting and opposing votes, respectively. The final verdict combines both the weighted scores and vote counts

$$\text{Verdict} = \begin{cases} \text{accept}, & \text{if } s_{\text{weighted}} > 0.5 \text{ and } n_{\text{sup}} \geq n_{\text{opp}} \\ \text{reject}, & \text{if } s_{\text{weighted}} < 0.3 \text{ or } \\ & n_{\text{opp}} > 1.5 \cdot n_{\text{sup}} \\ \text{uncertain}, & \text{otherwise.} \end{cases} \quad (30)$$

For accepted hypotheses, the final confidence score is computed as $s_{\text{final}} = s_{\text{weighted}} / \sum_a w_a$. The final output of AgentHOI is the set of accepted HOI instances

$$\mathcal{Y} = \{(\mathbf{b}_h, \mathbf{b}_o, c_o, v, s_{\text{final}}, \mathcal{A}) \mid h \in \mathcal{H}, \text{Verdict}(\mathbf{h}) = \text{accept}\} \quad (31)$$

where $\mathcal{A}$ contains the complete debate history, providing interpretable reasoning traces for each detection.

IV. EXPERIMENTS

A. Dataset and Evaluation Metrics

We evaluate our method on HICO-DET [32] and V-COCO [39], which are the most widely used benchmarks for HOI detection. HICO-DET contains 47776 images (38118 for training and 9658 for testing) with annotations for 600 HOI categories, composed of 80 object categories from COCO [40] and 117 verb categories. For evaluation metrics, we adopt the standard mean average precision (mAP) metric for evaluation.

B. Implementation Details

We use YOLOv8 as the default object detector, which provides a good balance between speed and accuracy. The detection confidence threshold is set to 0.25. For the HP, we employ two BLIP [41] models. The SEA uses Qwen2.5 [42] for generating interaction rules and commonsense reasoning. To ensure efficiency, we maintain a cache of LLM responses for all verb-object pairs encountered during evaluation. For verb-object pairs not in the cache, we fall back to predefined

TABLE I
PERFORMANCE COMPARISON ON HICO-DET TEST SET

Method	Backbone	Training-free	Data volume	mAP
<i>Two-stage Methods</i>				
HO-RCNN [32]	CaffeNet	✗	All	7.81
InteractNet [43]	ResNet-50-FPN	✗	All	9.94
GPNN [44]	ResNet-101	✗	All	13.11
iCAN [33]	ResNet-50	✗	All	14.84
<i>One-stage Methods (Transformer-based)</i>				
UnionDet [45]	ResNet-50-FPN	✗	All	17.58
IP-Net [46]	Hourglass-104	✗	All	19.56
HRNet [47]	ResNet-152	✗	All	21.93
PPDM-Hourglass [24]	Hourglass-104	✗	All	21.94
HOI-Trans [21]	ResNet-50	✗	All	23.46
GG-Net [48]	Hourglass-104	✗	All	23.47
PST [49]	ResNet-50	✗	All	23.93
<i>Weakly+ supervised HOI Methods</i>				
Explanation-HOI* [50]	ResNeXt101	✗	(object, interaction)	10.63
MAX-HOI [51]	ResNet101	✗	(object, interaction)	16.14
PPR-FCN* [52]	ResNet-50	✗	(object, interaction)	17.55
PGBL [20]	ResNet-50	✗	(object, interaction)	22.89
<i>Weakly supervised HOI Methods</i>				
SCG* [19]	ResNet-50	✗	(interaction)	7.05
VLHOI [53]	ResNet-50	✗	(interaction)	8.38
<i>Agent-based Methods (Ours)</i>				
AgentHOI	-	✓	No	13.26

TABLE II
PERFORMANCE COMPARISON ON V-COCO TEST SET

Method	Type	mAP
AlignFormer	Weakly+	14.15
AgentHOI (Ours)	Agent-based	14.58

TABLE III
COMPARISON OF INTERPRETABILITY FEATURES
ACROSS DIFFERENT METHOD TYPES

Feature	Two-stage	One-stage	Weakly	AgentHOI
Reasoning trace	✗	✗	✗	✓
Evidence explanation	✗	✗	✗	✓
Modular diagnosis	✗	✗	✗	✓
Human-readable output	✗	✗	✗	✓

rules. All experiments are conducted on the $8 \times$ NVIDIA RTX A6000 GPU with 48 GB memory. The framework is implemented in PyTorch 2.0.1 and Python 3.8.20.

C. Main Results

Table I presents the comparison of AgentHOI with existing methods on HICO-DET. We categorize the compared methods into four groups: two-stage methods, one-stage (Transformer-based) methods, weakly+, and weakly supervised HOI methods.

As shown in Table I, AgentHOI achieves 13.26 mAP, which is competitive with early two-stage methods but still below recent Transformer-based approaches. This gap is expected, since AgentHOI relies on zero-shot inference from pretrained models and rule-based reasoning rather than large-scale supervised training or HOI-specific fine-tuning. Notably, it outperforms existing weakly supervised methods such as VLHOI (8.38 mAP), demonstrating the potential of the proposed framework. We emphasize that the main contribution

TABLE IV
ABLATION STUDY ON INDIVIDUAL AGENTS. EACH ROW SHOWS
THE PERFORMANCE WHEN THE CORRESPONDING
AGENT IS REMOVED FROM THE DEBATE

Configuration	mAP
Full AgentHOI	13.26
w/o Visual Analysis Agent	11.57
w/o Semantic Expert Agent	11.34
w/o Physics Auditor Agent	11.94
w/o Debate (single round)	13.04

TABLE V
EFFECT OF THE NUMBER OF DEBATE ROUNDS

Round 1	Round 2	Round 3	Round 4	Round 5
13.04	13.25	13.26	13.25	13.24

TABLE VI
EFFECT OF DIFFERENT AGENT WEIGHT
CONFIGURATIONS IN THE CA

w_{VAA}	w_{SEA}	w_{PAA}	mAP
0.33	0.33	0.33	13.25
0.50	0.25	0.25	12.45
0.35	0.35	0.30	13.26
0.25	0.50	0.25	13.25
0.25	0.25	0.50	13.10

of this work is not state-of-the-art performance, but the exploration of a new multi-agent paradigm for HOI detection. On V-COCO (Table II), our method achieves 14.67 mAP, surpassing the traditional weakly+ baseline (14.17 mAP). The relatively modest performance on V-COCO is mainly due to the high similarity among action categories and their limited intra-class variation. A key advantage of AgentHOI is its interpretability, as shown in Table III.

D. Ablation Studies

We conduct ablation studies to analyze the contribution of each component in AgentHOI.

1) *Individual Agents*: Table IV shows the performance when removing individual agents from the debate process. Among all components, removing SEA causes the largest performance drop (−1.92 mAP), highlighting the importance of semantic reasoning in understanding interactions. The removal of VAA leads to a 1.69-mAP decrease. While removing PAA has a relatively smaller impact (−1.32 mAP), it still contributes meaningfully to filtering out physically implausible interactions. Most notably, replacing the debate mechanism with single-round voting results in 13.04 mAP, which demonstrates the effectiveness of multiround deliberation in achieving accurate predictions.

2) *Debate Rounds*: Table V analyzes the impact of the number of debate rounds. Performance improves from 1 to 3 rounds, then plateaus. This suggests that three rounds of debate are sufficient for agents to exchange information and reach consensus.

TABLE VII
EFFECT OF CONSENSUS THRESHOLD θ_{cons}

$\theta_{\text{cons}} = 0.3$	$\theta_{\text{cons}} = 0.6$	$\theta_{\text{cons}} = 0.7$	$\theta_{\text{cons}} = 0.9$
13.04	13.05	13.26	13.26

TABLE VIII
PERFORMANCE COMPARISON OF DIFFERENT PROMPT
TEMPLATES ON HICO-DET DATASET

Prompt Template	mAP
“a photo of a person [verb] a [object]”	12.85
“a person [verb] a [object]”	13.26
“a human [verb] a [object]”	12.54
“[verb] a [object]”	11.50

3) *Agent Weights*: Table VI explores different weight configurations for the CA. The balanced configuration (0.35, 0.35, 0.30) performs best. Over-weighting any single agent leads to performance degradation, confirming the importance of multiperspective evaluation. The VAA and SEA provide the two primary sources of discriminative evidence for HOI recognition: direct visual grounding and commonsense semantic plausibility. In contrast, the PAA mainly acts as a consistency filter to suppress physically implausible predictions. Therefore, we assign slightly higher weights to VAA and SEA, while keeping PAA only marginally lower.

4) *Consensus Threshold*: Table VII analyzes the impact of the consensus threshold θ_{cons} . A threshold of 0.7 achieves the best performance. Lower thresholds lead to premature consensus with insufficient deliberation, while higher thresholds may cause over-debate and noise accumulation.

5) *Impact of Different Prompt Templates*: As shown in Table VIII, we evaluate the impact of different prompt templates on the HICO-DET dataset. The template “a person [verb] a [object]” achieves the best performance with an mAP of 13.26, outperforming other variants. Compared to the commonly used template “a photo of a person [verb] a [object],” removing the prefix “a photo of” yields a 0.41-mAP improvement, suggesting that simpler and more direct descriptions better align with the visual semantics of HOIs. Replacing “person” with “human” leads to a performance drop of 0.72 mAP, indicating that the word “person” provides more effective semantic grounding in the vision-language embedding space. Furthermore, completely omitting the subject (i.e., “[verb] a [object]”) results in the lowest mAP of 11.50 mAP, demonstrating that explicitly mentioning the human subject is crucial for HOI detection. These results highlight the importance of prompt engineering in vision-language models, where subtle linguistic variations can significantly affect downstream task performance.

E. Error Analysis

The failure modes of AgentHOI on HICO-DET categorize them into four types: 1) object detection failures, where missed or misclassified small objects lead to incorrect HOI predictions; 2) spatial reasoning errors caused by inaccurate

bounding boxes under occlusion, misleading the interaction reasoning agents; 3) semantic ambiguity among visually similar interactions that are hard to distinguish from static images; and 4) rare interaction failures due to insufficient common-sense support for infrequent verb-object pairs. For each type, we also discuss potential optimization directions to mitigate these issues.

F. Discussion

1) *Advantages, Limitations and Future Work:* AgentHOI offers clear advantages over traditional methods, including strong interpretability through complete reasoning traces, flexible modular upgrades, and zero-shot robustness to rare HOI categories by leveraging pretrained knowledge, while opening a new paradigm for HOI detection. However, it still has several limitations: its performance remains below fully supervised state-of-the-art methods due to the lack of end-to-end training; the multi-agent debate process incurs a higher computational cost; and its effectiveness is constrained by object detection quality and manually designed rules. Future work will focus on developing more autonomous agents, integrating learnable modules, adopting stronger vision-language models, designing more efficient debate mechanisms, and extending the framework to related visual tasks. In particular, AgentHOI can zero-shot generate human-object interaction triplets and further combine them with an LLM for image description generation.

2) *Scalability:* The modular architecture of AgentHOI naturally supports the integration of new specialized agents. For example, a Pose Estimation Agent could be added to analyze human body keypoints for verifying pose-dependent interactions (e.g., “sit,” “bend”). To integrate a new agent, one only needs to: 1) implement its analysis function following the (stance, score, evidence) interface; 2) register it with the DM; and 3) assign a weight in the CA, which can be tuned on a validation set.

V. CONCLUSION

In this article, we present AgentHOI, a novel multi-agent debate framework for HOI detection. To the best of our knowledge, this is an early attempt to apply AI agent technologies to this task, offering a new paradigm beyond traditional end-to-end learning. Our framework introduces six collaborative agents that analyze interaction hypotheses from visual, semantic, and physical perspectives through a “hypothesis-debate-consensus” workflow. Experiments on HICO-DET demonstrate competitive performance with interpretable reasoning traces, highlighting unique advantages in interpretability and zero-shot generalization.

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